Type of Submission

Assignment for the lecture "E8069 Computational Macroeconomics."

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Problem 1: Question 1: The model, pen and paper

A household chooses consumption and assets to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = z_t + Ra_t$$

for every $t \geq 0$, with a_0 given. The gross interest rate $R = 1 + r$ is constant, u is strictly increasing, strictly concave, and differentiable, and income z_t is a Markov process. We write $\mathbb{E}[\cdot \mid z]$ for the expectation over z' given today's income. A prime denotes tomorrow, derivatives carry the variable as a subscript, $u_c(c)$ and $v_a(a, z)$.

Question 1.1: Optimality conditions — 10 points

(a) Write down the Bellman equation with a' as the only choice variable. Why is (a, z) the state?

(b) Derive the first-order condition for a'.

(c) Derive the envelope condition for $v_a(a, z)$ by plugging the policy $a' = g(a, z)$ into the Bellman equation and differentiating with respect to a, rather than quoting the general result.

(d) Combine the two into the Euler equation

$$u_c(c) = \beta R \, \mathbb{E}[u_c(c') \mid z].$$

Question 1.2: The analytical household of Acharya and Dogra — 25 points

Now fix the functional forms. Utility is

$$u(c) = -\frac{1}{\gamma} e^{-\gamma c}$$

with $\gamma > 0$, which is defined for every real c, there is no borrowing limit, $a' \in \mathbb{R}$, and income is normal and persistent,

$$z = \bar{z} + e, \qquad e' = \rho e + \sigma_e \varepsilon', \qquad \varepsilon' \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \qquad \rho \in [0, 1). \tag{1}$$

The unconditional standard deviation of income is

$$\sigma = \frac{\sigma_e}{\sqrt{1 - \rho^2}}.$$

The state is cash on hand $x = z + Ra$, the resources available today, together with the income deviation e. Conjecture that consumption is affine in both,

$$c = \mu x + \nu e + f,$$

with constants μ, ν, f to be determined.

(a) Write the Euler equation for CARA utility. Use the conjecture to derive the asset policy it implies, $a' = x - c$, and express tomorrow's consumption through x, e, and ε'. You should arrive at

$$\begin{aligned} c' = {} & \mu R(1-\mu)x \\ & + [\mu(\rho - R\nu) + \nu\rho]e \\ & + \mu(\bar{z} - Rf) + f \\ & + (\mu + \nu)\sigma_e \varepsilon'. \end{aligned} \tag{2}$$

It can be shown that, conditional on today's state (x, e), c' is normally distributed: its mean is the sum of the first four terms of (2), its variance is $(\mu + \nu)^2 \sigma_e^2$.

(b) If Y is normal, then

$$\mathbb{E}[e^Y] = \exp\left(\mathbb{E}[Y] + \frac{\mathrm{Var}(Y)}{2}\right).$$

Use this to evaluate the expectation in the Euler equation, take logs, and match the coefficients on x, e, and the constants to obtain

$$\begin{aligned} \mu &= \frac{r}{R}, \\ \nu &= \frac{\rho\mu}{R - \rho}, \\ f &= \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu + \nu)^2 \sigma_e^2}{2r}. \end{aligned} \tag{3}$$

(c) Give an economic interpretation of μ, and explain what its value r/R means for the household's behaviour.

(d) An innovation $\sigma_e \varepsilon$ raises consumption through e and through x. Show that consumption responds with

$$\mu + \nu = \frac{r}{R - \rho}$$

per unit of the innovation, the marginal propensity to consume out of an income surprise. How does it change with the persistence ρ, and what is the economic intuition behind its values in the limits $\rho \to 0$ and $\rho \to 1$?

(e) Write down the policy for i.i.d. income, $\rho = 0$.

Question 1.1

The household solves

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$\begin{aligned} c_t + a_{t+1} &= z_t + Ra_t, \\ a_0 &\text{ given}, \\ R &= 1 + r, \end{aligned}$$

where z_t follows a Markov process.

(a) Bellman equation

Using the budget constraint,

$$c_t = z_t + Ra_t - a_{t+1},$$

we can substitute consumption into the utility function:

$$u(c_t) = u(z_t + Ra_t - a_{t+1}).$$

The Bellman equation is therefore

$$\boxed{V(a, z) = \max_{a'} \left\{ u(z + Ra - a') + \beta \mathbb{E}\left[V(a', z') \mid z \right] \right\}.}$$

We now consider both a and z as the required state variables, as they jointly determine the agent's available current income for consumption.

While a is an endogenous state determined by the agent's past decisions, z is exogenously determined by the stochastic income process described as a Markov process.

(b) First-order condition

The first-order condition with respect to a' is

$$\begin{aligned} \text{FOC}_{a'}: \quad & -u_c(c) + \beta \mathbb{E}\left[V_a(a', z') \mid z \right] = 0 \\ \Leftrightarrow \quad & \boxed{u_c(c) = \beta \mathbb{E}\left[V_a(a', z') \mid z \right]} \end{aligned}$$

(c) Envelope condition

To derive the Envelope condition we plug in the policy function for $a' = g(a, z)$ into the Bellman equation to derive:

$$V(a, z) = u\big(z + Ra - g(a, z)\big) + \beta \mathbb{E}\left[V(g(a, z), z') \mid z \right].$$

Differentiating with respect to a and rearranging yields

$$\begin{aligned}
&V_a(a,z) = u_c(c)\left[R - g_a(a,z)\right] + \beta\,\mathbb{E}\left[V_a(g(a,z),z')g_a(a,z) \mid z\right] \\
\Leftrightarrow\; &V_a(a,z) = u_c(c)\left[R - g_a(a,z)\right] + \beta\,\mathbb{E}\left[V_a(g(a,z),z') \mid z\right] g_a(a,z) \\
\Leftrightarrow\; &V_a(a,z) = u_c(c)R - u_c(c)g_a(a,z) + \beta\,\mathbb{E}\left[V_a(g(a,z),z') \mid z\right] g_a(a,z) \\
\Leftrightarrow\; &V_a(a,z) = u_c(c)R + g_a(a,z)\left[\beta\,\mathbb{E}\left[V_a(g(a,z),z') \mid z\right] - u_c(c)\right]
\end{aligned}$$

From the first-order condition of the original Bellman equation, we know that

$$u_c(c) = \beta\mathbb{E}\left[V_a(a',z') \mid z\right].$$

Therefore, the term with $g_a(a,z)$ is equal zero and the equation simplifies to

$$V_a(a,z) = u_c(c)R + g_a(a,z)\underbrace{\left\{\beta\mathbb{E}\left[V_a(g(a,z),z') \mid z\right] - u_c(c)\right\}}_{=0}.$$

Thus, the envelope condition is

$$\boxed{V_a(a,z) = R\,u_c(c).}$$

(d) Euler equation

Iterating one period forward, we can plug the envelope condition into the first-order condition to derive the consumption Euler equation.

$$\begin{aligned}
&u_c(c) = \beta\,\mathbb{E}\left[V_a(a',z') \mid z\right] \\
&s.t.\; V_a(a',z') = R\,u_c(c') \\
\Leftrightarrow\; &u_c(c) = \beta\,\mathbb{E}\left[R\,u_c(c') \mid z\right] \\
\Leftrightarrow\; &\boxed{u_c(c) = \beta R\,\mathbb{E}\left[u_c(c') \mid z\right]}
\end{aligned}$$

Question 1.2

(a) Euler equation and asset policy

The Euler equation is derived as

$$\begin{aligned}
&u_c(c) = \beta R\,\mathbb{E}\left[u_c(c') \mid x,e\right] \\
&s.t.\; u_c(c) = -\frac{1}{\gamma}(-\gamma)e^{-\gamma c} = e^{-\gamma c} \\
\Leftrightarrow\; &\boxed{e^{-\gamma c} = \beta R\,\mathbb{E}\left[e^{-\gamma c'} \mid x,e\right].}
\end{aligned}$$

Starting with the asset policy, we use the conjecture to rewrite the budget constraint as

$$\begin{aligned} & c + a' = z + Ra \\ s.t.\ & x = z + Ra \\ & c = \mu x + \nu e + f \\ \Leftrightarrow\ & x = a' + \mu x + \nu e + f, \\ \Leftrightarrow\ & a' = x - (\mu x + \nu e + f) \\ \Leftrightarrow\ & \boxed{a' = x - c = g(x, e)} \end{aligned}$$

Returning to consumption tomorrow, we iterate the conjecture one period forward and substitute the expressions provided for x' and e':

$$\begin{aligned} & c' = \mu x' + \nu e' + f \\ s.t.\ & x' = z' + Ra', \\ & e' = \rho e + \sigma_e \varepsilon' \\ \Leftrightarrow\ & c' = \mu(z' + Ra') + \nu(\rho e + \sigma_e \varepsilon') + f \end{aligned}$$

However, we also know the expressions for z' and a' and plug them into the equation:

$$\begin{aligned} s.t.\ & z' = \bar{z} + e', \\ & e' = \rho e + \sigma_e \varepsilon', \\ & a' = x - \mu x - \nu e - f \\ \Leftrightarrow\ & c' = \mu\left[(\bar{z} + \rho e + \sigma_e \varepsilon') + R(x - \mu x - \nu e - f)\right] + \nu(\rho e + \sigma_e \varepsilon') + f \end{aligned}$$

Expanding and rearranging the equation yields the consumption policy function:

$$\begin{aligned} & \Leftrightarrow c' = \mu\bar{z} + \mu\rho e + \mu\sigma_e \varepsilon' + \mu R(1-\mu)x - \mu R\nu e - \mu R f + \nu\rho e + \nu\sigma_e \varepsilon' + f \\ & \Leftrightarrow c' = \mu R(1-\mu)x + [\mu\rho - \mu R\nu + \nu\rho]e + \mu(\bar{z} - Rf) + f + (\mu + \nu)\sigma_e \varepsilon' \\ & \Leftrightarrow \boxed{c' = \mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e + \mu(\bar{z} - Rf) + f + (\mu + \nu)\sigma_e \varepsilon'} \end{aligned}$$

(b) Evaluating the expectation

Turning back to our Euler equation, if c' is normally distributed, we can apply log-linearization on the expectation to yield

$$\mathbb{E}\left[e^{-\gamma c'} \mid x, e\right] = \exp\left(\mathbb{E}[-\gamma c' \mid x, e] + \frac{1}{2}\operatorname{Var}(-\gamma c' \mid x, e)\right)$$

$$\begin{aligned}
\text{s.t.} \quad \mathbb{E}[-\gamma c' \mid x, e] &= -\gamma\, \mathbb{E}[c' \mid x, e] \\
&= -\gamma\Big[\mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e \\
&\qquad + \mu(\bar{z} - Rf) + f + (\mu+\nu)\sigma_e \underbrace{\mathbb{E}[\varepsilon']}_{=0}\Big]
\end{aligned}$$

$$\begin{aligned}
\operatorname{Var}(-\gamma c' \mid x, e) &= \gamma^2 \operatorname{Var}(c' \mid x, e) \\
&= \gamma^2(\mu+\nu)^2\sigma_e^2
\end{aligned}$$

$$\begin{aligned}
\implies \quad \log\left(\mathbb{E}\left[e^{-\gamma c'} \mid x, e\right]\right) &= -\gamma\Big[\mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e \\
&\qquad + \mu(\bar{z} - Rf) + f\Big] \\
&\quad + \frac{1}{2}\gamma^2(\mu+\nu)^2\sigma_e^2.
\end{aligned}$$

Taking logs of the Euler equation and plugging in our result for the log of the expectation yields

$$\begin{aligned}
-\gamma c &= \ln(\beta R) - \gamma\Big[\mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e \\
&\qquad + \mu(\bar{z} - Rf) + f\Big] + \frac{1}{2}\gamma^2(\mu+\nu)^2\sigma_e^2 \\
\Leftrightarrow \quad c &= -\frac{1}{\gamma}\ln(\beta R) + \mu R(1-\mu)x + [\mu(\rho - R\nu) + \nu\rho]e \\
&\qquad + \mu(\bar{z} - Rf) + f - \frac{1}{2}\gamma(\mu+\nu)^2\sigma_e^2
\end{aligned}$$

Since this equation has to hold with the original conjecture, we can match the coefficients.

Coefficient μ Matching the coefficient on x, assuming $\mu \neq 0$:

$$\begin{aligned}
& \mu = \mu R(1-\mu) \\
\Leftrightarrow \quad & 1 = R(1-\mu) \\
\Leftrightarrow \quad & \mu = 1 - \frac{1}{R} \\
& \quad = \frac{R-1}{R}.
\end{aligned}$$

Since $R = 1 + r$, we obtain

$$\boxed{\mu = \frac{r}{R}}.$$

Coefficient ν Matching the coefficient on e:

$$\begin{aligned} & \nu = \mu(\rho - R\nu) + \nu\rho \\ \Leftrightarrow \quad & \nu = \mu\rho - \mu R\nu + \nu\rho \\ \Leftrightarrow \quad \nu(1 + \mu R - \rho) &= \mu\rho \\ \Leftrightarrow \quad & \nu = \frac{\mu\rho}{1 + \mu R - \rho}. \end{aligned}$$

Substituting $\mu R = r = R - 1$ yields

$$\boxed{\nu = \frac{\mu\rho}{R - \rho}}.$$

Constant f Matching the constant terms:

$$\begin{aligned} & f = \mu(\bar{z} - Rf) + f - \frac{1}{2}\gamma(\mu + \nu)^2\sigma_e^2 - \frac{1}{\gamma}\ln(\beta R) \\ \Leftrightarrow \mu R f &= \mu\bar{z} - \frac{1}{2}\gamma(\mu + \nu)^2\sigma_e^2 - \frac{1}{\gamma}\ln(\beta R) \\ \Leftrightarrow f &= \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma\mu R} - \frac{\gamma(\mu + \nu)^2\sigma_e^2}{2\mu R} \\ s.t.\ \mu &= \frac{r}{R} \\ \Leftrightarrow f &= \frac{\bar{z}}{R} - \frac{\ln(\beta R)R}{\gamma r R} - \frac{\gamma(\mu + \nu)^2\sigma_e^2 R}{2rR} \\ \Leftrightarrow f &= \boxed{\frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu + \nu)^2\sigma_e^2}{2r}} \end{aligned}$$

(c) Economic interpretation

The coefficient μ describes the share of cash on hand consumed by the household in any period. Its value $\mu = \frac{r}{R}$ determines that households only consume the annuity value of their cash on hand and therefore represents the marginal propensity to consume.

(d) Consumption response to an innovation

The response of consumption to an income innovation is

$$
\begin{aligned}
\frac{\partial c}{\partial(\sigma_e\varepsilon)} &= \frac{\partial c}{\partial x}\frac{\partial x}{\partial z}\frac{\partial z}{\partial e}\frac{\partial e}{\partial(\sigma_e\varepsilon)} + \frac{\partial c}{\partial e}\frac{\partial e}{\partial(\sigma_e\varepsilon)} \\
&= \mu + \nu \\
s.t.\ \mu &= \frac{r}{R} \\
\nu &= \frac{\rho\mu}{R-\rho} \\
\Leftrightarrow \frac{\partial c}{\partial(\sigma_e\varepsilon)} &= \frac{r}{R} + \frac{\rho\mu}{R-\rho} \\
s.t.\ \mu &= \frac{r}{R} \\
\Leftrightarrow \frac{\partial c}{\partial(\sigma_e\varepsilon)} &= \frac{r}{R} + \frac{\rho r}{R(R-\rho)} \\
&= \frac{r(R-\rho)}{R(R-\rho)} + \frac{\rho r}{R(R-\rho)} \\
&= \frac{rR - r\rho + r\rho}{R(R-\rho)} \\
&= \frac{rR}{R(R-\rho)} \\
&= \frac{r}{R-\rho} \\
\Leftrightarrow \frac{\partial c}{\partial(\sigma_e\varepsilon)} &= \boxed{r(R-\rho)^{-1}} \\
\Leftrightarrow \frac{\partial c}{\partial(\sigma_e\varepsilon)} &= \boxed{\mu + \nu = \frac{r}{R-\rho}}
\end{aligned}
$$

Therefore, to derive the effect of the shock persistence on consumption change to an innovation shock is

$$\frac{\partial}{\partial\rho}\left(\frac{r}{R-\rho}\right) = \frac{r}{(R-\rho)^2}.$$

For $\rho \to 0$, consumption increases by the present value of the shock, as the household will save most of the income shock as assets.

For $\rho \to 1$, the income shock permanently increases labor income, and the household updates its consumption to its new permanent income, as $\frac{\partial c}{\partial(\sigma_e\varepsilon)} = 1$.

(e) Consumption policy for i.i.d. shocks

For i.i.d. shocks, we substitute $\rho = 0$ into our conjecture and our derived coefficients μ, ν, and f and simplify:

$$
\begin{aligned}
& c = \mu x + \nu e + f \\
s.t.\ & \mu = \frac{r}{R} \\
& \nu = \frac{\mu\rho}{R-\rho} \\
& f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_e^2}{2r} \\
\Leftrightarrow\ & c = \frac{r}{R}x + \frac{\mu\rho}{R-\rho}e + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_e^2}{2r} \\
& s.t.\ \rho = 0 \\
\Leftrightarrow\ & c = \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_e^2}{2r}
\end{aligned}
$$

However, from subquestion d, we know that for μ and ν the following relationship holds:

$$\mu + \nu = \frac{r}{R-\rho}$$

For $\rho = 0$, we plug this expression into our policy function and yield:

$$
\begin{aligned}
c &= \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+\nu)^2\sigma_e^2}{2r} \\
s.t.\ \mu+\nu &= \frac{r}{R} \\
\Leftrightarrow c &= \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma\left(\frac{r}{R}\right)^2\sigma_e^2}{2r} \\
\Leftrightarrow c &= \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma r^2\sigma_e^2}{2rR^2} \\
\Leftrightarrow c &= \boxed{\frac{rx+\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma\, r\sigma_e^2}{2R^2}}
\end{aligned}
$$

Problem 2: Question 2: The model, on a computer: the CARA household

Solve the household of Question 1.2 with $\gamma = 2$, $r = 0.04$, $\beta = 1/R$, $\bar{z} = 1$, $\sigma = 0.1$, $\rho = 0.9$. The notebook discretizes z by Rouwenhorst's method on $N_z = 7$ points and uses $N_a = 400$ equidistant asset grid points on $[-15, 25]$.

The two value function iterations stop when the value moves by less than $\varepsilon = 10^{-6}$, the three policy methods when the policy moves by less than $\varepsilon_{\text{pol}} = 10^{-6}$, all with an iteration cap of 5000 as a safety net.

Errors against the exact policy are reported as e_W, the largest absolute error in consumption units over the grid points with $a_i \in [0, 8]$, the window inside which the grid bounds are not felt, and e_G, the same maximum over the whole grid, both computed by the notebook.

Question 2.1: The exact policy of the discretized problem **4 points**

On the computer, income lives on the N_z points of a Rouwenhorst chain instead of the normal distribution. This approximation induces a small difference between the analytical policy of Part 1 and the exact policy of the problem the computer solves,

$$c_{im}^{\text{exact}} = \mu x_{im} + \nu e_m + f_{N_z}.$$

The notebook computes μ, ν, and f_{N_z} for the chain. Compare them with the values that the formulas of Question 1.2 imply.
By how much do they differ, and is this an economically meaningful difference?

Question 2.1:

When we compare the exact coefficients implied by the closed form solution with the computed coefficients in table 1, we can observe that they are identical at least up to the third decimal digit. The largest reported difference is for ν, but is smaller than $\frac{1}{1000}$. Therefore, the approximation deviation introduced by the Rouwenhorst approximation is not economically meaningful for the CARA model simulated here.

Table 1: Comparison of Analytical and Discretized CARA Policies

Parameter	Gaussian closed form	Discretized policy
μ	0.038462	0.038462
ν	0.247253	0.247189
f	0.95766091	0.95766213
Consumption difference at extreme income states: 1.7×10^{-5}		

Problem 3: Question 3: The model, on a computer: the CRRA household

Solve a household with CRRA utility,

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

with $\gamma = 2$, income

$$\ln z' = \rho \ln z + \sigma_e \varepsilon'$$

with $\rho = 0.9$ and $\sigma_e = 0.2$, discretized by Rouwenhorst's method on $N_z = 7$ points, a borrowing limit $a' \geq 0$, $r = 0.03$, and $\beta = 0.96$.
There is no exact policy: the Euler equation errors are only the check. The solvers are the ones of Part 2. What changes is the model, and you only change how it enters the solvers.

Question 3.1: From one model to the other **8 points**

Compare the CARA and the CRRA models on the following points, in a few sentences each.

(a) The utility function: u, u_c, u_c^{-1}, and the values of c for which they are defined.

(b) The income process: what the chain discretizes and how income in levels is obtained from it.

(c) The asset grid: lower bound, upper bound and what pins it down, and where the grid points of the discretization should be dense.

(d) The borrowing constraint, and the admissible consumption at a state (a_i, z_m) that follows from it and the grid.

Question 3.1.: Changes introduced by CRRA

When we simulate a CRRA model instead of an CARA model, the following functions are changed:

- u: $\underset{c\in\mathbb{R}\setminus\{0\}}{u(c)} = \frac{c^{1-\gamma}}{1-\gamma}$
- u_c: $\underset{c\in\mathbb{R}\setminus\{0\}}{u_c(c)} = c^{-\gamma}$
- u_c^{-1}: $\underset{m\in\mathbb{R}\setminus\{0\}}{u_c^{-1}(m)} = m^{-\frac{1}{\gamma}}$
- Income process:
  - Process: $\ln z' = \rho \ln z + \sigma_e \varepsilon'$
  - Discretization: The Rouwenhorst method remains unchanged. However, to derive the level from the log income values from the income process, we have to reverse the logarithmization
- Asset grid:
  - Lower bound: As the CRRA model includes a borrowing constraint, the new lower bound is equal to this borrowing constraint, here zero.
  - Upper bound: There is no definitive value for the upper bound. However, the upper bound of the asset grid must be beyond what households can reach in the stationary wealth distribution as otherwise the optimization problem would be faced with an inequality in the Euler equation as well. By trial and error, the upper bound should be chosen such that all methods report that the computed asset policy function should never or virtually never reach the upper bound.
  - Discretization: Due to the introduction of the borrowing constraint, households can reach states where the constraint is binding, Therefore, we need to allocate more grid points closer to the lower bound of the asset grid, which we here do with quadratic pacing.
- The borrowing constraint and the upper limit for admissible consumption choices remain unchanged in the CRRA model compared to the CARA model. However, CRRA utility modifies the lower bound of feasible consumption. As the CRRA utility function does not allow for zero consumption and as negative consumption is infeasible, we restrict the lower bound of the set to $\max(x_i - a_{max}, 10^{-10})$ to circumvent this issue.

For the utility function, its derivative and the inverse of said derivative, the function is mathematically defined for all consumption and income levels except for zero. However, we should exclude negative income values as they are not economically feasible.